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Raman spectroscopic study of gadolinium(III) in sodium-aluminoborosilicate glasses

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Abstract

A Raman spectroscopy study was performed on a series of sodium-aluminoborosilicate glasses with Gd_2O_3 from 0 mass% up to its solubility (47 mass% or 13.58 mol%). Experimentally measured spectra were fitted with a Gaussian function for each individual band without any restriction on the band position, width, and intensity. The evolution of the spectroscopic bands appears to result from *partitioning* of the rare earth cation, as its dissolution mechanism, in these borosilicate glasses. Specifically, the evolution of the Raman bands was correlated well with Gd cations *partitioning* in the borate-rich environment at low Gd_2O_3 concentration, $\text{Gd}_2\text{O}_3/[\text{1/3B}_2\text{O}_3] < 1$, and excess Gd cations, $\text{Gd}_2\text{O}_3-1/3\text{B}_2\text{O}_3$, in the silicate-rich environment at high Gd_2O_3 concentration, $\text{Gd}_2\text{O}_3/[\text{1/3B}_2\text{O}_3] > 1$. Raman bands near 1420 and 710 cm^{-1} suggest the presence of a local Gd-metaborate environment, which appeared to occur at all Gd_2O_3 concentrations. The bands near 300 and 910 cm^{-1} further suggest the formation of Gd–O–Gd clusters in the silicate-rich environment at high Gd_2O_3 concentrations. © 2001 Elsevier Science B.V. All rights reserved.

1. Introduction

There have been numerous spectroscopic studies of local environments of lanthanide elements (Ln) such as La(III), Nd(III), Gd(III), and Eu(III), in oxide systems of borate [1], aluminoborate [2–4], silicate [5–7], sodiumborosilicate [8–10], and sodium-aluminoborosilicate glasses [11,12]. In the borate systems, the results suggest that Ln elements dissolve in glass forming Ln-metaborate-like local structure [1–3] whereas in the silicate systems, they dissolve in the glass creating some

non-bridging oxygens [5,6]. Furthermore, formation of Ln–O–Ln clusters was reported for both borate [1] and silicate [5–7] systems. In the literature, however, Ln–O–Ln clustering phenomenon in borosilicate systems was not reported to the best of our knowledge though it would be expected.

The formation of glass can be generalized in terms of a reaction between the non-network-forming cations (basic component) and the network-forming groups (acidic component) according to Lewis theory. ¹ The local environment

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¹ Based on the Lewis theory, an acid is an atom, ion, radical, or molecule, which can form a chemical bond with its counterpart by accepting an electron pair. A base is an atom, radical, or molecule, which can donate an electron pair to form a chemical bond with its counterpart. Neutralization is the formation of a covalent bond though co-ordination by an acid and a base.

and/or geometrical stability of the non-network-forming cations may be governed by the strength of an electrostatic bond in terms of z/CN (where z is the charge and CN is the co-ordination number of the cation with its nearest oxygen) [13,14]. Recent studies [15–17] of silicate, borate, and borosilicate glasses using the state of the art of high-resolution nuclear magnetic resonance (NMR) spectroscopy show and/or suggest that borosilicate glasses may be reasonably treated as a mixture consisting of two network-forming groups, a borate-rich and a silicate-rich environment. The spatial dimension and degree of their unmixing may be of the order of a couple of nanometers or less in the quenched glasses [6]. Unmixing of the borate-rich and silicate-rich environments at a sub-nanoscale in quenched sodium-borosilicate glasses quenched with different rates was recently reported [18].

An understanding of the unmixing phenomenon is important for chemically designing and/or tailoring glass systems for future optical, electronic, and magnetic applications. In applications of nuclear materials vitrification, another study has shown that trivalent plutonium, Pu(III), is much more soluble in alkali-aluminoborosilicate glass than Pu(IV) [19]. The same solution behavior was also observed between Ce(III) and Ce(IV) in alkali-aluminoborosilicate glasses [20]. Therefore, Ln(III) may be used as a surrogate to study the local environment of trivalent actinide elements, such as Pu(III), in glass, which is also important for the development of borosilicate glass host matrices for chemical stabilization of radioactive materials.

Recently, we showed that the dissolution behavior of Ln(III), i.e., La(III) and Gd(III), is similar in a series of sodium-aluminoborosilicate glasses with a mole ratio of $\text{Na}_2\text{O}/(\text{Na}_2\text{O} + \text{Al}_2\text{O}_3)$

from 0 to 1 [21–25]. The solubility limit of La_2O_3 or Gd_2O_3 in the glasses was controlled by the formation of a dispersed amorphous phase rich in boron and La and Gd cations over the composition region where $-0.5 < (\text{Na}_2\text{O}-\text{Al}_2\text{O}_3)/\text{B}_2\text{O}_3 < 0.5$ [21,22]. On either side of $|\text{Na}_2\text{O}-\text{Al}_2\text{O}_3|/\text{B}_2\text{O}_3 > 0.5$, no microphase separation was observed; the solubility limit of La_2O_3 or Gd_2O_3 was controlled by the crystallization of La- or Gd-silicate phases [21–25]. In this paper, we report a Raman spectroscopy study of a previously reported system with $(\text{Na}_2\text{O}-\text{Al}_2\text{O}_3)/\text{B}_2\text{O}_3 = 0.75$. The results detail Gd(III) dissolution in the glass, which supports our previously proposed partitioning mechanism of Ln(III) for explaining how Ln cations dissolve in the borosilicate system. Furthermore, formation of Gd–O–Gd clusters in the silicate-rich environment was demonstrated.

2. Experimental

A single sodium-aluminoborosilicate glass, B20 (Table 1), was used as a baseline in our Raman study. We systematically introduced Gd_2O_3 at 5, 20, 30, and 47 mass%. Table 1 shows composition of the five glasses with and without Gd_2O_3 . Samples were melted at 1550 °C for 1 h and then 1450 °C for 1 h followed by immersing bottom portion of the platinum crucible in water to quench the glass from 1450 °C as previously reported [19]. At 48 mass% Gd_2O_3 , crystallization of $\text{NaGd}_9(\text{SiO}_4)_6\text{O}_2$ occurred in the glass [25]. No Raman study was performed with crystallized glasses. However, we did perform Raman measurements on a pure $\text{NaGd}_9(\text{SiO}_4)_6\text{O}_2$ phase synthesized by using a conventional solid state reaction route at 1600 °C.

Table 1
Target compositions (mol%) for B20 glasses with and without Gd_2O_3

Glass ID	B20	B20Gd5	B20Gd20	B20Gd30	B20Gd47
Al_2O_3	4.76	4.72	4.15	4.43	4.12
B_2O_3	19.05	18.87	16.60	17.70	16.46
SiO_2	57.14	56.61	49.81	53.11	49.38
Na_2O	19.05	18.87	16.60	17.70	16.46
Gd_2O_3	0.00	0.92	3.86	7.06	13.58

Raman spectroscopic measurements were performed on the monolithic glass samples. The Raman spectra were collected with a Triple Raman Spectrometer with the 488 nm line of an argon ion laser for excitation. The power at the sample was approximately 200 mW. An LN/CCD detector was used with an exposure time of 1–10 s and a slit width of 100 μm . Raman spectra were obtained at 180° scattering geometry on glass sample plates. The Raman peak positions were measured based on the reference of the known bands of aspirin with an accuracy within $\pm 2\text{ cm}^{-1}$. Raman spectra from glass samples were collected over the region from 150 to 1500 cm^{-1} . Spectral analyses were performed with Galactic Industries GRAMS/32 software.

3. Results

Fig. 1 shows Raman spectra of the B20 glasses with Gd_2O_3 from 0 up to the solubility limit of 13.58 mol% (47 mass%). Several distinct features can be seen from the spectra. First, bands near 700 and 1420 cm^{-1} formed when Gd_2O_3 is doped in the baseline was observed. Second, the position of a band near 1050 cm^{-1} shifted to lower wave numbers as Gd_2O_3 concentration in glass was increased. Third, there is a significant reduction in

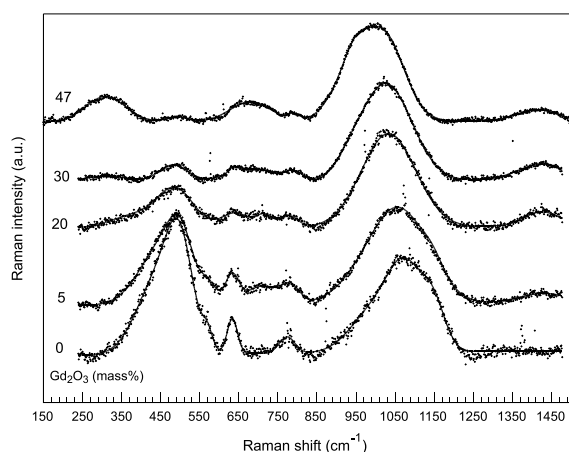


Fig. 1. Experimental (full circle) and fitted (line) Raman spectra of sodium-aluminoborosilicate glasses with and without Gd_2O_3 .

the intensity of the band near 450 cm^{-1} as Gd_2O_3 concentration in the glass was increased. Finally, a band near 300 cm^{-1} formed for the glasses with 7.06 and 13.58 mol% (or 30 and 47 mass%) Gd_2O_3 .

To further elucidate a change in the glass network structure, which is associated with Gd cation dissolution in the glass, we used a Gaussian function to fit each spectrum. For the baseline, eight individual bands were used in the initial fitting and the results showed that all of them were necessary for a good fit. For the Gd-bearing glasses, the initial fit started with 10 individual bands; the two additional bands were based on our observation of the two new bands, near 1400 and 700 cm^{-1} . However, depending on the concentration of Gd_2O_3 in the glass, the final fit was accomplished with 10 or fewer bands. For the entire curve fitting, no restrictions were applied to the position, width, and intensity of each individual band. The resultant fitted spectra are shown in Fig. 1. Overall, the fit to the observed spectra is good.

For each glass, Fig. 2 shows the individual bands and positions that were used in the fit to the observed spectra. Fig. 3 shows some of prominent band positions versus Gd_2O_3 concentration. For the band near 1420 cm^{-1} , the band position decreases as Gd_2O_3 concentration increases (Fig. 3(a)). This is also the case for the band near 1050 cm^{-1} (Fig. 3(b)). For the band near 780 cm^{-1} , its position appeared to be stepwise, higher at higher Gd_2O_3 concentration, which possibly represents two different structural environments, rather than a change within a single environment. The band near 700 cm^{-1} appears to be also stepwise, but lower at higher Gd_2O_3 concentration, which, again, suggests two different structural environments. The observed stepwise change for the glasses with Gd_2O_3 between 3.86 and 7.06 mol% could be associated with the difference between Gd cations dissolving into two different environments as will be discussed later. Finally, for the band near 500 cm^{-1} , there seems to be a minimum in its position for the glass with 7.06 mol% Gd_2O_3 (30 mass%). In the following discussion, we will first discuss possible mechanisms for the network structure changes associated with the Gd-doping and then the fitted bands will be discussed based on the proposed mechanisms.

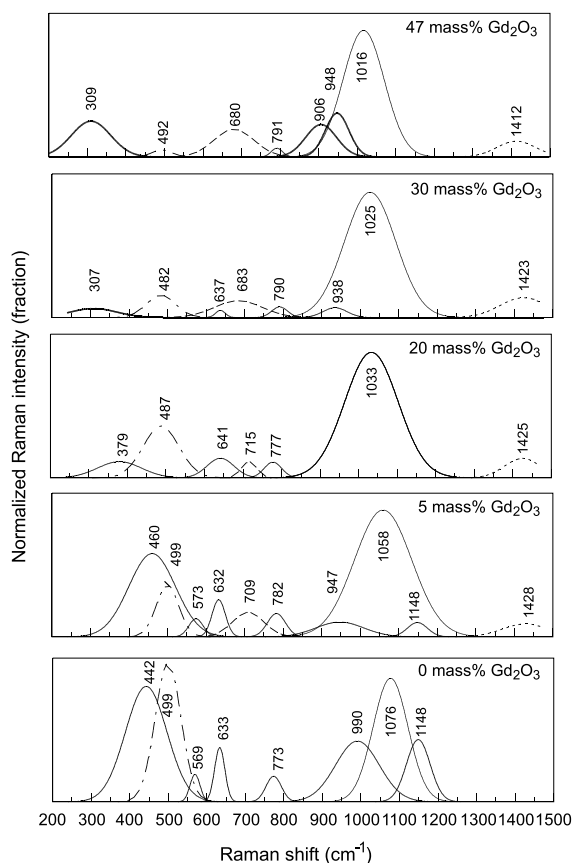


Fig. 2. Raman curve fitting results (individual bands and band positions).

4. Discussion

4.1. Mechanism of Gd dissolution in borosilicate glass

Considering the existence of unmixing, at a sub-nano- or nanoscale, between borate-rich and silicate-rich environments, a *partitioning* mechanism can be used to explain the speciation of borate groups, $\text{BO}_3 \leftrightarrow \text{BO}_4$, in borosilicate systems in which monovalent [15–17], divalent [26,27], and trivalent cations [8,9] are introduced. The *partitioning* of the network-modifying cations in the borate-rich environment can be understood in terms of an acid–base reaction. Optical basicity is in the order $\text{B}_2\text{O}_3 < \text{SiO}_2 < \text{Al}_2\text{O}_3$ [28]. Accordingly, *partitioning* of the network-modifying cations in the borate-rich environment of the

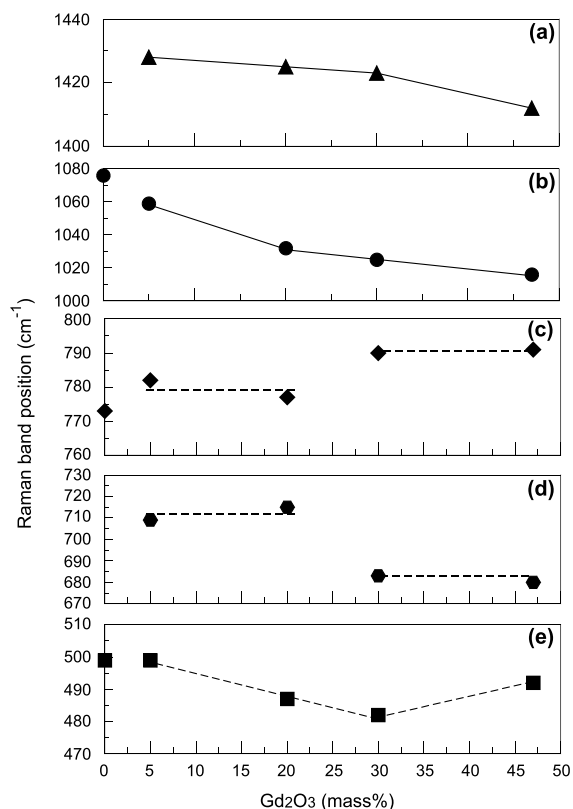


Fig. 3. Raman band position versus Gd_2O_3 concentration in glass.

borosilicate system is preferred over the silicate-rich environment. However, it must be noted that Al_2O_3 should be considered as a strong acid if its reaction with alkali metal oxides is interpreted in terms of an acid–base one. The partitioning mechanism has been also applied to understanding microphase development in Gd-doped sodium-aluminoborosilicate glasses with $|\text{Na}_2\text{O}-\text{Al}_2\text{O}_3|/\text{B}_2\text{O}_3$ less than 0.5 [29]. Specifically, we treated the system as a mixture of borate-rich and silicate-rich environments based on previously reported studies [9,10,15–17], in which Gd cations preferentially partition to the borate-rich environment until all available borate groups are consumed by Gd cations. Although excess sodium ($\text{Na}_2\text{O}-\text{Al}_2\text{O}_3$) cations react with B_2O_3 forming BO_4 based on the literature data as well as that shown by our Raman data for the B20 baseline glass (Fig. 1), Raman spectra of the Gd-bearing glasses (Fig. 1)

suggest that Gd cations are favorably competing with Na cations for the borate-rich environment. This competition results in the redistribution of the ‘debonded’ or ‘repelled’ Na cations in the silicate-rich environments to achieve their local charge neutrality (more discussion on this finding will be given later). Therefore, for the glasses in this study, we assume that the excess B_2O_3 available to Gd cations is equal to its total concentration in each glass with Gd_2O_3 . The local environment of (Gd, B)-rich environment ought to resemble that of Gd-metaborate, $1BO_4:1Gd:2BO_3$ [1–4]. Under the local Gd-metaborate structure hypothesis, the excess Gd cations, $[Gd_2O_3]_{ex} = Gd_2O_3 - 1/3B_2O_3$, dissolve in the silicate-rich environment followed by crystallization of the Gd-silicate phase.

Partitioning of Ln(III) in the sodium-aluminum-borosilicate glasses is affected by other glass constituents according to phonon side band (PSB) study of Eu(III) in the $Na_2O-B_2O_3-SiO_2$ systems investigated by Gatterer et al. [10]. For the glasses with Na_2O/B_2O_3 ratio from 0 to about 0.3 and SiO_2/B_2O_3 from 0.3 to 6, Eu cations were shown to partition in the borate-rich environments to greater than 95%. On the other hand, the concentration or a percentage of total Eu cations in the silicate-rich portion of the glass increased, not exceeding 50% total Eu, as Na_2O/B_2O_3 increased above 0.3. The amount of Eu cations dissolved in the silicate-rich environment was further shown to be affected by SiO_2/B_2O_3 [10].

For the glasses with a given SiO_2/B_2O_3 and Na_2O/B_2O_3 less than 0.3, the previous NMR studies [15–17] showed that nearly all Na cations reacted with boron or partitioning into the borate-rich environment. The concentration of BO_4 linearly increased from zero concentration to a maximum of about 50% of the total boron oxide as Na cation concentration was increased, suggesting that no Na cations dissolved in the silicate-rich environment. In pure silica network, solubility of Ln(III) cations is very low, a few parts per million, and elevated Ln(III) cations in pure silica glass tend to form clusters as demonstrated by the fluorescence lifetime study of Nd-doped SiO_2 by Namikawa et al. [30] and molecular dynamic simulations comparing local La–O environments in Na_2O-SiO_2 and SiO_2 glasses by Park et al. [31].

Therefore, without any network-modifying cations such as Na cations, the dissolution of Eu cations in silicate-rich environment is not favorable. At the same time, there is sufficient amount of borate groups for dissolving Eu cations in this case and hence, redistribution of the ‘repelled’ Na cations into the silicate network (as we proposed based on this work) is not expected. For Na_2O/B_2O_3 greater than 0.3, the NMR studies [15–17] showed that BO_4 decreased and BO_3 increased as a result of the formation of other borate groups in borate glass systems. However, in borosilicate systems, there exists a plateau over an extended Na_2O/B_2O_3 greater than 0.3, which implies the dissolution of excess Na cations in the silicate-rich environment. It is known that Na cations in the silicate-rich environments create non-bridging oxygen environment that seems to favor an increase in Eu solubility based on the reported Eu partitioning in the silicate-rich environment at Na_2O/B_2O_3 values greater than 0.3. A similar partitioning phenomenon was also reported for Nd cations in these systems [8].

A competition among the glass network-forming oxides for Na_2O can be ranked in the order $Al_2O_3 > B_2O_3 > SiO_2$, according to several spectroscopic studies [17,32–34]. For the systems under current investigation, the value of $(Na_2O-Al_2O_3)/B_2O_3$ is 0.75 and SiO_2/B_2O_3 is approximately 3. By analogy to the Eu study for the host glass composition of similar values of Na_2O/B_2O_3 and SiO_2/B_2O_3 , approximately 10% of the total Gd cations may be expected in the silicate-rich environments for our glasses with 0.92 and 3.86 mol% Gd_2O_3 , at which concentrations, a threshold for saturating the borate-rich environment, i.e., $Gd_2O_3 - 1/3[B_2O_3] = 0$, has not been reached. For the glasses with 7.06 and 13.58 mol% Gd_2O_3 , the excess Gd cations, as represented by $Gd_2O_3 - 1/3[B_2O_3]$, are expected to partition exclusively to the silicate-rich environment. This hypothesis is indirectly supported by the fact that only a Gd-silicate phase crystallized from the melt above the Gd_2O_3 solubility limit [21] and by our recent optical spectroscopic study of Nd_2O_3 in $Al_2O_3-B_2O_3-SiO_2$ glasses [11]. Unlike the change in borate speciation reaction caused by alkali metal cations [15], the borate-rich environment

bonded with Gd cations is expected to remain unchanged and resembles the Gd-metaborate structure of $1\text{BO}_4:1\text{Gd}:2\text{BO}_3$. The high stability of the Gd-metaborate local structure has been reported for aluminoborate and borosilicate glasses with high rare earth concentrations such as Y_2O_3 in $\text{Al}_2\text{O}_3\text{--B}_2\text{O}_3$ by Chakraborty et al. [2], La_2O_3 in $\text{Al}_2\text{O}_3\text{--B}_2\text{O}_3$ by Brow and Tallant [3,4], Gd_2O_3 in $\text{Na}_2\text{O--Al}_2\text{O}_3\text{--B}_2\text{O}_3\text{--SiO}_2$ [12], and Nd_2O_3 in $\text{Al}_2\text{O}_3\text{--B}_2\text{O}_3\text{--SiO}_2$ [11].

In the B20Gd5 and B20Gd20 glasses, $\text{Gd}_2\text{O}_3/(1/3\text{B}_2\text{O}_3)$ is 0.15 and 0.70, respectively. According to the proposed partitioning mechanism for Gd cation dissolution, we expect that Gd cations for the B20Gd5 and B20Gd20 glasses partitioned primarily to the borate-rich environment. In the B2030 and B20Gd47 glasses, the $\text{Gd}_2\text{O}_3/(1/3\text{B}_2\text{O}_3)$ is 1.20 and 2.48, respectively. Hence, about 15% and 60% of the total Gd_2O_3 (1.16 and 8.10 mol% Gd_2O_3) are expected to be in the silicate-rich environments in the two glasses. In the following section, Raman bands will be discussed in detail within the context of the proposed partitioning mechanism and the 1:3 local environment for Gd cations in the borate-rich groups.

4.2. Local structural environments of borate-rich and silicate-rich groups

Raman bands for borate and aluminoborate glasses with and without rare earth elements have been studied by Meera and Ramakrishna [35], Terashima et al. [1] and Brow and Tallant [3,4]. Extensive Raman spectroscopic studies on silicates can be found in the literature such as the work by McMillan [36] and Mysen [37]. Raman work on silicates doped with Ln(III) by Ellison and Hess [5] and Schaller et al. [6] was also reported. The Raman band assignments based on the previous studies are summarized in Table 2 along with band positions for the spectra obtained from this study, which were determined from curve fitting. Based on our proposed partitioning mechanism for Ln(III) dissolution in borosilicate glasses and band assignments from the literature, a detailed discussion of the band assignments follows.

For the B20 baseline glass, a band over $1300\text{--}1600\text{ cm}^{-1}$ is absent indicating a negligible amount of trigonal BO_3 units in the glass. This implies that Na cations in the glass stabilize a majority of boron in a tetrahedral configuration, BO_4 , and is supported by NMR studies [15–17]. On the other hand, a new band near 1420 cm^{-1} is seen in all Gd-doped glasses; the band position moves to lower wave numbers as Gd_2O_3 concentration increases. The new band is associated with the BO_3 units (Table 2). Therefore, Gd cations are likely dissolved in the borate-rich environment at least at low concentrations. At the same time, another new band can be seen at 709 and/or 715 cm^{-1} for the B20Gd5 and B20Gd20 glasses, respectively. The band positions are assigned to the vibration of the metaborate structure (Table 2). Therefore, the formation of two new bands, one near 1420 cm^{-1} and another near 710 cm^{-1} , in the two low Gd-doped glasses can be used as an evidence that indeed Gd cations dissolved in the borate-rich environment and their local structure resembles $1\text{BO}_4:1\text{Gd}:2\text{BO}_3$ as found in Gd-metaborate phase [1]. Further indication of Gd cation partitioning in the borate-rich environment can be found by examining the 1420 cm^{-1} band intensity. As shown in Fig. 2, its intensity initially increases with Gd, between 0.92 and 3.86 mol%, and then levels off at higher Gd concentrations, between 7.06 and 13.58 mol%. The latter suggests Gd saturation in the borate-rich environment as will be discussed later.

The partitioning mechanism is also substantiated by the following findings. For the B20Gd30 with 3.86 mol% Gd_2O_3 and B20Gd47 with 13.58 mol% Gd_2O_3 , a new band near 300 cm^{-1} is observed. The band suggests the formation of Gd–O–Gd cluster environment as reported for both borate and silicate glasses that contain elevated concentration of rare earth elements [1,5]. At the same time, a new band near 906 cm^{-1} is observed in the B20Gd47 glass. Our previous X-ray diffraction study showed that the B20Gd48 glass contained Gd-silicate crystals and the Raman spectrum of this pure Gd-silicate phase exhibits a very broad band near 900 cm^{-1} and a strong band near 850 cm^{-1} as well as an intermediate band near 310 cm^{-1} , as shown in Fig. 4. Accordingly, the bands near 300 and 906 cm^{-1} for the glasses,

Table 2

Band assignments of borate and silicate environments reported in the literature and determined from this study

Reported position (cm ⁻¹)	Assignments	Observed position (cm ⁻¹)
<i>Borate environments with and without rare earth doping</i>		
300	Ln (La, Nd, Sm, Pr)–O stretch (Ln–O–Ln clusters)	
450–570	B–O–B stretch, BO ₄ units	569–573
480–500	Al–O ⁻ or Al–O–B, aluminate network	499–482
600–650	B–O–B, ring metaborate units	632–641
700–730	B–O–B, chain metaborate units	709–715
700–720	Al–O ⁻ or Al–O–B, aluminate network	
750–780	B–O–B, rings with BO ₄ units	(773–791)
820–850	B–O–B, pyroborate units	
890–940	B–O ⁻ , orthoborate units	
900–1000	B–O–B, BO ₄ units	
960–980	Al ^[4] –O–B ^[3] , aluminoborate network	
1075–1150	B–O ⁻ and B–O–B, diborate units	
1200–1300	B–O ⁻ , pyroborate units	
1300–1600	B–O ⁻ , chain and ring metaborate units	1412–1428
<i>Silicate environments with and without rare earth doping</i>		
222–230		N/A
304–312	Ln(La, Nd, Gd)–O stretch (Ln–O–Ln clusters)	307–309
		379
430–440	Q ⁴ units	442–460
524–530	Q ³ units	
592–612	Q ² units	
600	Q ³ units mixed bending and stretching with Q ⁴	
640	Q ² units mixed bending and stretching with Q ⁴	
670	Q ¹ units mixed bending and stretching with Q ⁴	
700	Q ¹ units	680–683
760–782	SiO ₄ cage	(773–791)
850	Q ⁰ units	
900	Q ¹ units	
940–950	Q ² units with Ln (La, Gd)	938–948
950–1000	Q ² units	
1000–1030	Q ³ units with Ln (La, Gd)	1016–1033
1050–1100	Q ³ units	1058–1076
1200, 1150	Q ⁴ units	1148

Note: Q⁴ = SiO₂, Q³ = Si₂O₅²⁻, Q² = SiO₃²⁻, Q¹ = Si₂O₇⁶⁻, and Q⁰ = SiO₄⁴⁻.

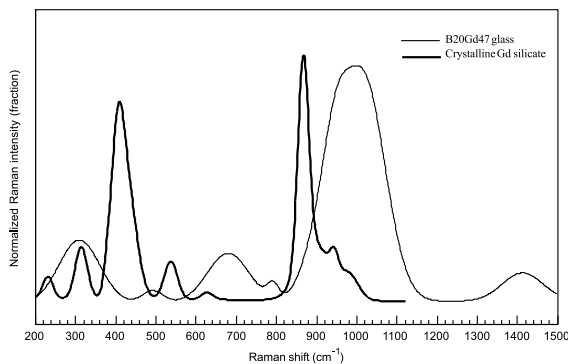


Fig. 4. Raman spectra of glass containing 13.58 mol% Gd₂O₃ (47 mass%) and pure Gd-silicate crystalline phase.

B20Gd30 and B20Gd47, with excess 1.16 and 8.10 mol% Gd₂O₃, may represent different vibration modes of the Gd–O–Gd clusters in the Gd-silicate groups [5]. The Gd–O–Gd clusters are known to exist in crystalline Gd-silicates [38]. Hence, the formation of the bands near 300 and 900 cm⁻¹ in the two high Gd-doped glasses may be used as an additional evidence that at low Gd concentration, i.e., Gd₂O₃/(1/3B₂O₃) = 0.15 or 0.70, respectively, Gd cations primarily partition in the borate-rich environment. In this case, some Gd cations, less than 10% of total Gd (0.09 and 0.39 mol% Gd₂O₃ for B20Gd5 and B20Gd20, respectively), are expected to be in the silicate-rich en-

vironment [10] and formation of the Gd–O–Gd clusters in the silicate-rich environment is, therefore, unlikely. At high Gd concentrations, on the other hand, the saturation of the borate groups, i.e., $\text{Gd}_2\text{O}_3/(1/3\text{B}_2\text{O}_3) = 1.20$ or 2.48 , respectively, forces excess Gd cations to dissolve in the silicate-rich environments. Approximately 15% and 60% of total Gd_2O_3 (1.16 and 8.10 mol% Gd_2O_3 , respectively) are expected to dissolve in the silicate-rich environments in the two high-Gd glasses, respectively. Hence, Gd–O–Gd clusters form in the silicate-rich environments as suggested by a new band near 300 cm^{-1} (Fig. 2). Although the band could also be used as an indicator of Gd–O–Gd clusters in the borate-rich environment (Table 2), it is ruled out for its absence in the glasses with 0.92 and 3.86 mol% Gd_2O_3 that are supposed to be completely dissolved in the borate-rich environment.

The band near 990 cm^{-1} in the B20 baseline glass may consist of both BO_4 and Q^2 units, both of which bond with Na cations depending on the Gd_2O_3 concentration. For the silicate-rich environments, addition of Gd would result in more Q^2 groups [6], while the band decreases sharply in the glass with 0.92 mol% Gd_2O_3 in the B20Gd5 glass. Therefore, the 990 cm^{-1} band may arise primarily from BO_4 units otherwise its intensity is expected to increase if the band is primarily from Q^2 in silicate-rich environment due to the reaction between Gd cations and O–Si–O. As will be discussed later regarding the effect of Gd dissolution on the bands at 1076 and 1150 cm^{-1} for the B20 baseline glass without Gd_2O_3 , the observed sharp decrease in the 990 cm^{-1} band intensity likely result from the competition between Gd and Na cations for the borate groups. It is also consistent with the proposed local Gd-metaborate-like structure, i.e., for every Gd cation would require one BO_4 and two BO_3 units.

For the B20Gd30 and B20Gd47 glasses, the bands at 938 and 940 cm^{-1} , respectively, can be assigned to the silicate groups within which every silicate tetrahedron is bonded with two Gd cations, i.e., forming two non-bridging oxygens per SiO_4 or $\text{Q}_{[\text{Gd}]}^2$. The results from the curve fitting for the B20Gd5 and B20Gd20 glass with $\text{Gd}_2\text{O}_3/$

$(1/3[\text{B}_2\text{O}_3]) < 1$ suggest the absence of the $\text{Q}_{[\text{Gd}]}^2$. The absence of $\text{Q}_{[\text{Gd}]}^2$ also implies that few Gd cations are dissolved in the silicate-rich environments in the B20Gd5 and B20Gd20 glasses, which is consistent with our proposed Gd partitioning mechanism.

The band at 1076 cm^{-1} in the B20 glass can be assigned to the silicate units of Q^3 type (Table 2). However, some of the Q^3 units may come from the silicate units bonded with Gd cations, $\text{Q}_{[\text{Gd}]}^3$, in the Gd-doped glasses. Less than 10% of the total Gd cations are expected in the B20Gd5 and B20Gd20 glasses according to the study of the Eu-doped $\text{Na}_2\text{O–B}_2\text{O}_3\text{–SiO}_2$ glasses [10]. Hence, the bands at 1058 and 1033 cm^{-1} are still considered to be Q^3 in which non-bridging oxygens of the Si–O tetrahedra are bonded primarily with Na cations; the band shift implies that some should be $\text{Q}_{[\text{Gd}]}^3$ although there is no clear separation of the two bands from Q^3 and $\text{Q}_{[\text{Gd}]}^3$. In a study of $\text{K}_2\text{O–SiO}_2\text{–La}_2\text{O}_3$ glasses, it was shown that non-bridging oxygen of Si–O tetrahedra bonded to La cations, $\text{Q}_{[\text{La}]}^3$, had a different environment than non-bridging oxygen of Si–O tetrahedra bonded to K cations, Q^3 , based on their distinct Raman bands [6]. In our study, such a distinction was not observed from the experimental Raman spectra, which may be contributed to a better mixing of the rare earth elements in the silicate-rich network containing AlO_4 units as compared to that in the silicate-rich environment without AlO_4 as reported in the literature [6]. The band position further decreases to 1025 cm^{-1} for the B20Gd30 and then to 1016 cm^{-1} for the B20Gd47 glass. For the two glasses, the ratio of $\text{Gd}_2\text{O}_3/(1/3\text{B}_2\text{O}_3)$ is 1.20 and 2.48, respectively, suggesting that the borate groups have completely reacted. Hence, more $\text{Q}_{[\text{Gd}]}^3$ are expected to form in the silicate-rich environment, since the excess Gd cations ($\text{Gd}_2\text{O}_3 - 1/3\text{B}_2\text{O}_3$) dissolve in the silicate-rich environment according to the proposed Gd cation partitioning mechanism.

The band near 1150 cm^{-1} from the B20 glass can be related to Q^4 groups. In the B20Gd5 glass, the addition of Gd_2O_3 decreases the band intensity significantly. It is possible that the reduction in Q^4 results from an increase in Na cations in the silicate-rich environment. As shown in Fig. 1, the B20 glass

without Gd contains boron mostly in tetrahedral configuration, BO_4 that must be stabilized by Na cations since there are enough Na cations, i.e., $(\text{Na}_2\text{O}_3 - \text{Al}_2\text{O}_3)/\text{B}_2\text{O}_3 = 0.75$. With the addition of Gd, the simultaneous formation of trigonal BO_3 units and the decrease in Q^4 units are possible when the following conditions are satisfied:

1. Gd cations compete with Na cations in the borate-rich environment and
2. the Na cations that are excluded from the borate-sites by Gd cations are then stabilized only in the silicate-rich environment by reacting with silicate units, Q^4 , forming Q^3 units.²

The resulting increase in Q^3 , suggested by the band intensity increase, comparing the B20 with B20Gd5 and B20Gd20, can be attributed to the redistribution of Na cations from the borate-rich environment to the silicate-rich environment. In addition, up to 10% of total Gd cations in the silicate-rich environment can also result in a Q^3 increase at the expense of Q^4 in the B20Gd5 glass. However, the band at 1058 cm^{-1} is within the range of band positions found in silicates, suggesting that the former dominate the $\text{Q}^4 \leftrightarrow \text{Q}^3$ speciation reaction for the low Gd-doped glasses.

5. Conclusions

Raman spectra of a single baseline sodium-aluminoborosilicate glass with Gd_2O_3 concentration from 0 to 13.58 mol% were studied in detail by examining their deconvoluted spectra in comparison with assignments reported in the literature for borate and silicate glasses with and without rare earth elements. The bands derived from this study are further discussed within the context of rare earth element *partitioning* as a dissolution mechanism in borosilicate glasses. Our study shows that the Raman band evolution at different Gd_2O_3 concentrations can be explained with our proposed Gd cation *partitioning* mechanism. Gadolinium

first partitions to the borate-rich environment at low Gd concentrations [$\text{Gd}_2\text{O}_3 / (1/3\text{B}_2\text{O}_3) < 1$] and then to the silicate-rich environments at high Gd concentrations [$\text{Gd}_2\text{O}_3 / (1/3\text{B}_2\text{O}_3) > 1$]. Saturation of the borate groups with Gd leads to excess Gd cations, $\text{Gd}_2\text{O}_3 - 1/3\text{B}_2\text{O}_3$, which then dissolve in the silicate-rich environment until Gd-silicate precipitates. The local environment of Gd-metaborate-like structure, $1\text{BO}_4:1\text{Gd}:2\text{BO}_3$, was suggested by the Raman bands near 1420 and 710 cm^{-1} . Formation of Gd–O–Gd clusters in the silicate-rich environment was suggested by the Raman bands near 300 and 910 cm^{-1} when $\text{Gd}_2\text{O}_3 / (1/3\text{B}_2\text{O}_3) > 1$.

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² These Na cations are not likely stabilized by a charge coupling with some types of Gd–O anions like AlO_4^- since many previous studies [1–4,35] showed that Gd cations are stabilized in the borate-rich or silicate-rich environments without other network-modifying cations.

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